



TEST 2 FSPM0034 Question SEM3 2019 2020-I

Statistics and Probability (Universiti Teknologi Malaysia)



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FOUNDATION PROGRAMME

FSPM0034

Statistics and Probability

Semester 3 2019/2020

TEST 2(20%)

Duration: 90 minutes

INSTRUCTION: ANSWER ALL QUESTIONS AND SHOW ALL YOUR WORKINGS.

1. A computer store sells X computers a day with the following probability:

$$P(X = x) = \frac{6 - x}{20}; x = 0, 1, 2, 3, 4$$

(a) Calculate the average sales per day. [3 marks]

(b) Hence, find $P\left(|X - E(X)| < \frac{3}{2}\right)$. [3 marks]

2. Let X be the time in minutes that a vehicle has to wait at a traffic light. X has a cumulative density function given by;

$$F(x) = \begin{cases} 0; & x \leq 0 \\ 1 - k(10 - x)^2; & 0 \leq x \leq 10 \\ 1; & x \geq 10 \end{cases}$$

Where k is a constant.

(a) Determine the value of k [2 marks]

(b) Find $P(3 \leq X \leq 9)$ [2 marks]

(c) Determine the probability density function of x for $0 \leq x \leq 10$ [3 marks]

(d) Find the variance of X . [5 marks]

3. At the local cycling club, the expected number of people that can cycle 80 km per day is 4.5 and the variance is 3.15. Find the probability that at least three people can cycle 80 km per day.

[5 marks]

4. From a study conducted by a Research Association, it is known that 1 person out of every 4 is allergic to a particular pollen. A group of 20 foundation students volunteer to try out a new treatment.

(a) What is the expectation of the number of allergic people in the group? [2 marks]

(b) What is the probability that

(i) exactly two are allergic? [2 marks]

(ii) not more than two are allergic? [2 marks]

(c) How large a sample would be needed for the probability of it containing at least one allergic person to be greater than 99.9%? [5 marks]

(d) What assumptions have you made in your answer (c)? [2 marks]

5. On a lonely rural road in Northern Utah, cars are observed passing at the rate of 8 per day and lorries at the rate of 4 per day. On the road is an old cattle grid which will soon need repair. The local works department decide that if the probability of more than 2 vehicles per hour passing is less than 2% then the repairs to the cattle grid can wait until next spring, otherwise it will have to be repaired before the winter.

When will the cattle grid have to be repaired? [4 marks]

End